KEVIN GREIF (he/him)

Artificial Intelligence — Scientific Research — Statistical Methods

Physics Ph.D. candidate with 5+ years of experience developing and deploying ML solutions for complex scientific problems. Published researcher specializing in statistical ML, generative models including diffusion, and large-scale HPC deployments. Seeking AI research roles, particularly working on applications in the natural sciences or related fields.

EXPERIENCE

Ph.D. candidate in Physics - AI methods development **2020 - Present**
University of California, Irvine

- Developed novel **latent diffusion models** for inverse problems in particle physics, achieving state-of-the-art performance on high-dimensional scientific data
- Expertise in designing artificial intelligence systems for handling complex variable-dimensional data structures
- Published research in peer-reviewed conference proceedings (NeurIPS 2023, SciPost Physics 2024/2025)
- Presented research at international physics and machine learning conferences

Researcher at CERN - AI methods deployment **2020 - Present**
ATLAS Collaboration

- Trained thousands of large attention-based neural networks models on HPC resources
- Experience in developing and optimizing complicated neural network training workflows
- Benchmarked, optimized hyper-parameters, and quantified uncertainties for many neural network based classifiers. Improved performance by a factor of 5x over traditional methods
- Created and released open-source datasets and evaluation tools, facilitating reproducibility and community adoption

Undergraduate Research Assistant - Neural Network Architecture Design **2018 - 2020**
University of Notre Dame

- Designed physics-inspired neural network architectures incorporating domain knowledge, achieving improved performance over generic architectures

Teaching, Leadership, and Service

- **ATLAS Early Career Scientist Board (2024-Present)**: Advocate for 1000+ early career scientists within international scientific collaboration, organize career development workshops and networking events
- **Teaching Assistant (2019-2021)**: Taught and mentored 100+ students in physics and computational methods courses

SKILLS

Artificial Intelligence: Attention-based neural networks, diffusion models, hyperparameter optimization

Tools and infrastructure: PyTorch, PyTorch Lightning, TensorFlow, JAX, Git, Weights and Biases

Programming: Python, C++, Git, HPC resources and distributed computing, CI/CD pipelines, testing frameworks

Data Science: Large-scale data processing, statistical methods, data visualization

Research: Published author (NeurIPS, SciPost Physics), peer review, technical writing, scientific communication

International Collaboration: Active member of large (1000+ researchers) scientific collaboration, experience working in teams to produce high-quality research results

EDUCATION

University of California, Irvine **2020 - Present (Expected 2026)**
Ph.D. in Physics, Department of Physics and Astronomy — Advisor: Daniel Whiteson
Focus: Artificial Intelligence applications in particle physics research

University of Notre Dame **2016 - 2020**
B.S. in Physics and Program of Liberal Studies, Magna Cum Laude
Honors: Outstanding Physics Major Award, Departmental award for outstanding writing, Phi Beta Kappa

SELECTED PUBLICATIONS

I only list publications in which I played a significant role. For a full list of publications, see my Inspire HEP profile.

End-to-end latent variational diffusion models for inverse problems in high energy physics — *NeurIPS*

Proposes a novel latent variational diffusion model. Applied to solving an important inverse problem in particle physics, demonstrating then state-of-the-art performance for generative model based approaches to this problem.

Full event particle-level unfolding with variable-length latent variational diffusion — *SciPost Physics*

Proposes a latent variational diffusion model for solving variable dimensional inverse problems in particle physics. Main development was the ability to process variable-length feature spaces.

Generative unfolding of jets and their substructure — *arXiv:2510.19906*

Proposes a flow-matching model based method for solving a very high (100+) and variable-dimensional inverse problem in particle physics.

Explicit or Implicit? Encoding Physics at the Precision Frontier — *arXiv:2603.08802*

Benchmarks symmetry-equivariant (explicit) vs. large-scale pretrained (implicit) approaches to encoding prior physics priors in neural networks.

Accuracy versus precision in boosted top tagging with the ATLAS detector — *JINST*

Benchmarking of various methods for solving an important classification problem in particle physics. Also includes rigorous uncertainty quantification for the performance of the methods, and documents the release of a high-quality public dataset for the community.

The landscape of unfolding with machine learning — *SciPost Physics*

A community comparison study of unfolding methods in particle physics. I curated one of the comparison datasets used in this study.